

Date 6) Transfer of files from one Virtual Machine to another

Aim:

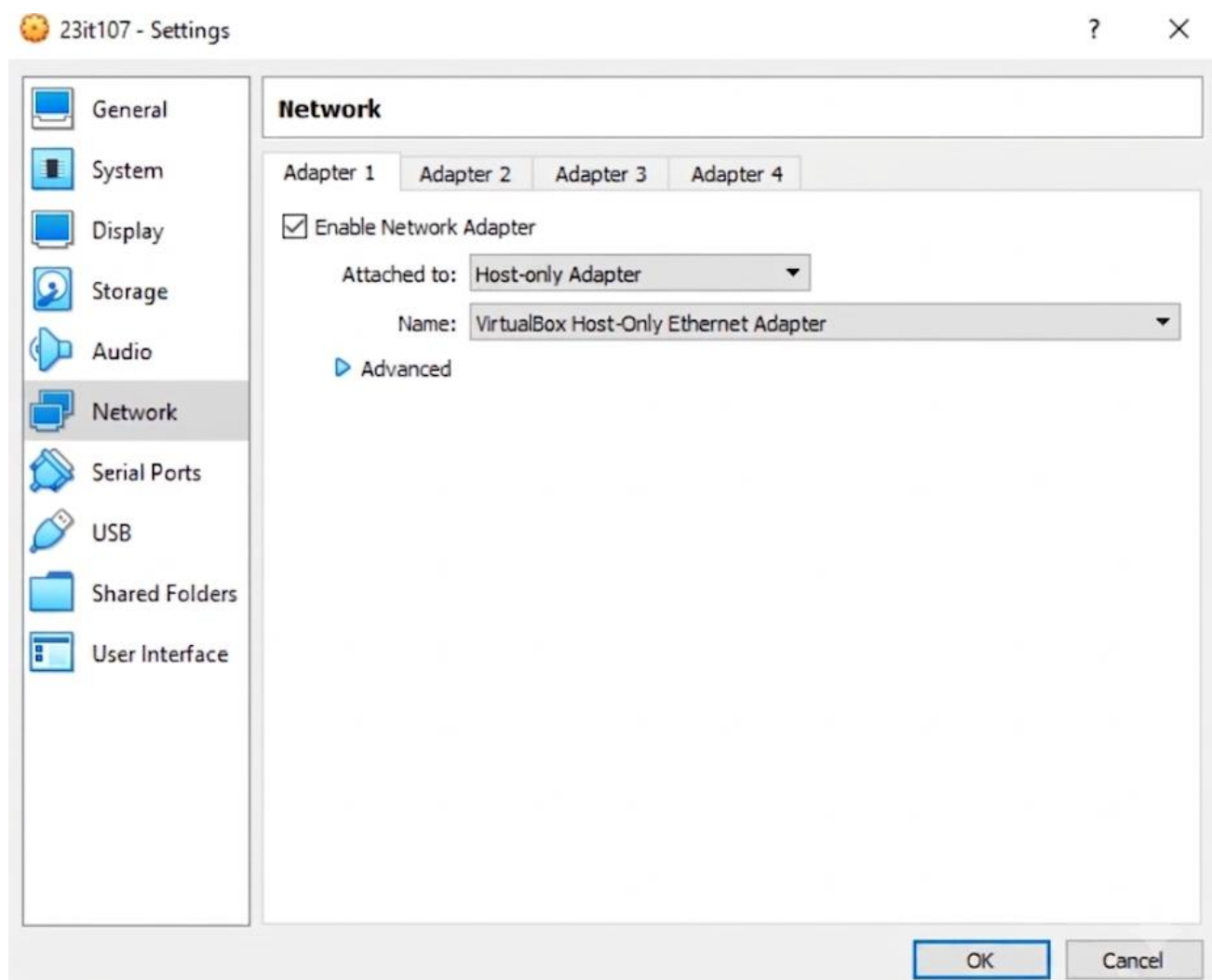
To transfer files from Ubuntu VM to Kali VM using NFS share, enabling read and write access on both ends.

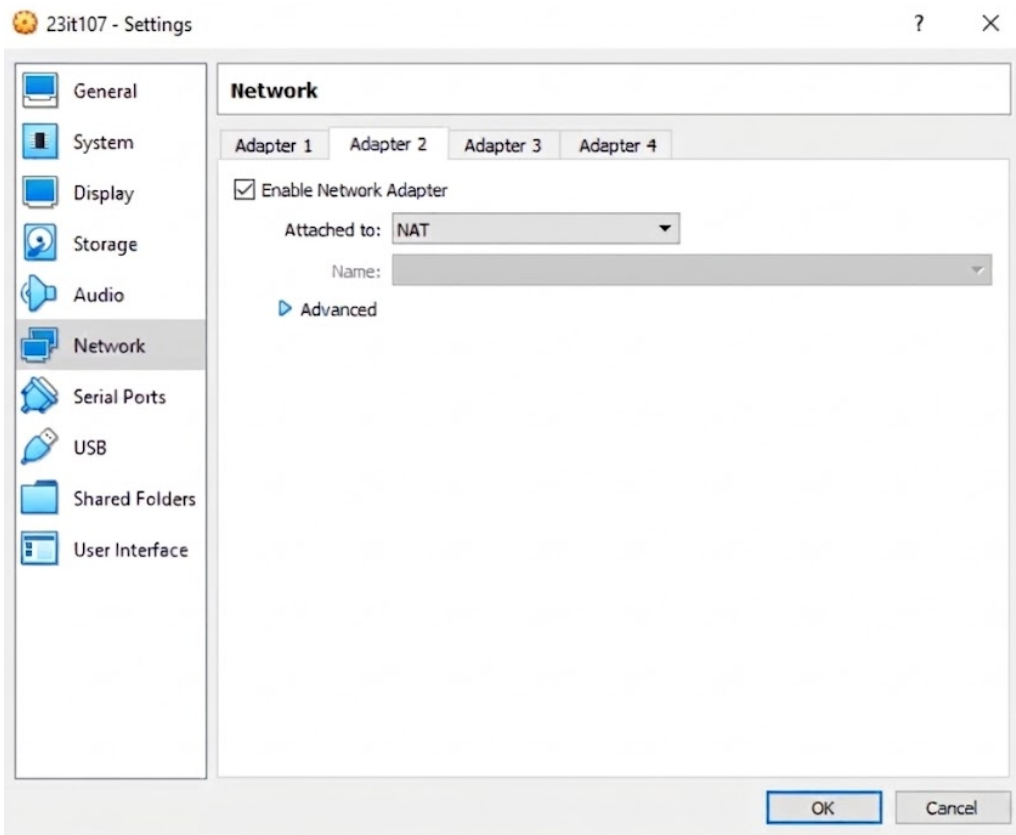
Steps:

1.Configure VirtualBox Network

Set both VMs' network adapter to "**Host-only Adapter**" in adapter 1 and **NAT** in adapter 2 (for installing packages) so they are on the same subnet.

Go to settings -> network -> adapter 1-> Host only adapter and Adapter2-> NAT





2. Check IP Addresses

On both ubuntu as well as kali **ip a** (check ip) Ensure both are in same subnet (eg 192.168.56.103/24 ubuntu and 192.168.56.104 kali) both in same subnet 192.168.56.0/24

```

File Machine View Input Devices Help
Activities Terminal Tue 14:18
tce@tce-VirtualBox: ~
File Edit View Search Terminal Help
inet 127.0.0.1/8 scope host lo
    valid_lft forever preferred_lft forever
inet6 ::1/128 scope host
    valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
    roup default qlen 1000
    link/ether 08:00:27:a1:92:af brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.101/24 brd 192.168.56.255 scope global dynamic noprefixrout
e enp0s3
    valid_lft 492sec preferred_lft 492sec
    inet6 fe80::b531:384e:a678:5eaa/64 scope link noprefixroute
    valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
    roup default qlen 1000
    link/ether 08:00:27:e6:04:b4 brd ff:ff:ff:ff:ff:ff
    inet 10.0.3.15/24 brd 10.0.3.255 scope global dynamic noprefixroute enp0s8
    valid_lft 86292sec preferred_lft 86292sec
    inet6 fe80::34c5:b304:970f:fe5b/64 scope link noprefixroute
    valid_lft forever preferred_lft forever
4: virbr0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOW
    N group default qlen 1000
    link/ether 52:54:00:2f:46:58 brd ff:ff:ff:ff:ff:ff
    inet 192.168.122.1/24 brd 192.168.122.255 scope global virbr0
    valid_lft forever preferred_lft forever
5: virbr0-nic: <BROADCAST,MULTICAST> mtu 1500 qdisc fq_codel master virbr0 stat
e DOWN group default qlen 1000
    link/ether 52:54:00:2f:46:58 brd ff:ff:ff:ff:ff:ff
tce@tce-VirtualBox:~$

```

Sudo dhclient eth0 – if ip address not assigned

```

tce@TCEDA: ~
File Actions Edit View Help
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group
   default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast sta
   te UP group default qlen 1000
   link/ether 08:00:27:a5:e9:c1 brd ff:ff:ff:ff:ff:ff
   inet 192.168.56.102/24 brd 192.168.56.255 scope global dynamic eth0
       valid_lft 593sec preferred_lft 593sec
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast sta
   te UP group default qlen 1000
   link/ether 08:00:27:96:e6:9c brd ff:ff:ff:ff:ff:ff
   inet 10.0.3.15/24 brd 10.0.3.255 scope global dynamic noprefixroute
   eth1
       valid_lft 84879sec preferred_lft 84879sec
   inet6 fe80::a00:27ff:fe96:e69c/64 scope link noprefixroute
       valid_lft forever preferred_lft forever

```

3. Set up NFS Server on Ubuntu VM(Here ubuntu act as server)**❖ Sudo apt update**

```

tce@tce-VirtualBox:~$ sudo apt update
[sudo] password for tce:
Get:1 https://dl.google.com/linux/chrome/deb stable InRelease [2,548 B]
Err:1 https://dl.google.com/linux/chrome/deb stable InRelease
  The following signatures couldn't be verified because the public key is not a
  vailable: NO_PUBKEY FD533C07C264648F
Hit:2 http://in.archive.ubuntu.com/ubuntu bionic InRelease
Get:3 http://security.ubuntu.com/ubuntu bionic-security InRelease [102 kB]
Get:4 http://in.archive.ubuntu.com/ubuntu bionic-updates InRelease [102 kB]
Get:5 http://in.archive.ubuntu.com/ubuntu bionic-backports InRelease [102 kB]
Get:6 http://security.ubuntu.com/ubuntu bionic-security/main amd64 DEP-11 Meta
ata [76.8 kB]
Get:7 http://in.archive.ubuntu.com/ubuntu bionic-updates/main amd64 DEP-11 Meta
data [297 kB]

```

❖ Update and install NFS server:

```
sudo apt update
```

```
sudo apt install nfs-kernel-server -y
```

```
tce@tce-VirtualBox:~$ sudo apt install nfs-kernel-server -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  keyutils libnfsidmap2 libtirpc1 nfs-common rpcbind
Suggested packages:
  open-iscsi watchdog
The following NEW packages will be installed:
  keyutils libnfsidmap2 libtirpc1 nfs-common nfs-kernel-server rpcbind
0 upgraded, 6 newly installed, 0 to remove and 58 not upgraded.
Need to get 492 kB of archives.
```

❖ Create folder to share:

sudo mkdir -p /srv/nfs/sharedfolder (or any folder name)

❖ Set permissions:

- sudo chown nobody:nogroup /srv/nfs/sharedfolder
- sudo chmod 777 /srv/nfs/sharedfolder

```
tce@tce-VirtualBox:~$ sudo mkdir -p /srv/nfs/sharedfolder
tce@tce-VirtualBox:~$ sudo chown nobody:nogroup /srv/nfs/sharedfolder
tce@tce-VirtualBox:~$ sudo chmod 777 /srv/nfs/sharedfolder
```

❖ Edit exports file to allow Kali IP access:

sudo nano /etc/exports (it open in nano editor)

Add /srv/nfs/sharedfolder < kali ip >(rw,sync,no_subtree_check,no_root_squash)

```
tce@tce-VirtualBox:~$ sudo nano /etc/exports
File Edit View Search Terminal Help
GNU nano 2.9.3 /etc/exports

# /etc/exports: the access control list for filesystems which may be exported
# to NFS clients.  See exports(5).
#
# Example for NFSv2 and NFSv3:
# /srv/homes hostname1(rw,sync,no_subtree_check) hostname2(ro,sync,no_subtree_check)
#
# Example for NFSv4:
# /srv/nfs4 gss/krb5i(rw,sync,fsid=0,crossmnt,no_subtree_check)
# /srv/nfs4/homes gss/krb5i(rw,sync,no_subtree_check)
#
/srv/nfs/sharedfolder 192.168.56.102(rw,sync,no_subtree_check,no_root_squash)
```

Click ctrl+o and enter and ctrl x (to save and came out of nano editor)

❖ Export and restart NFS server:

sudo exportfs -a

sudo systemctl restart nfs-kernel-server

```
tce@tce-VirtualBox:~$ sudo nano /etc/exports
tce@tce-VirtualBox:~$ sudo exportfs -a
tce@tce-VirtualBox:~$ sudo nano /etc/exports
tce@tce-VirtualBox:~$ sudo systemctl restart nfs-kernel-server
tce@tce-VirtualBox:~$
```

- ❖ Allow/disable the firewall for the kali linux (skipping shown already firewall in disable mode)

```
tce@tce-VirtualBox:~$ sudo ufw allow from 192.168.56.102 to any port nfs
[sudo] password for tce:
Skipping adding existing rule
```

4. Set up NFS Client on Kali VM

- ❖ Install NFS client:

- sudo apt update
- sudo apt install nfs-common -y

(here apt update not work because of network is in off so go to wired connection top right corner change the wired connection(NAT) and after above two command change again the wired connection1(Host only adapter)

```
(tce@TCEDA)-[~]
$ sudo apt update
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Err:1 http://kali.download/kali kali-rolling InRelease
  The following signatures couldn't be verified because the public key is
  not available: NO_PUBKEY ED65462EC8D5E4C5
Reading package lists... Done
W: GPG error: http://kali.download/kali kali-rolling InRelease: The foll
  owing signatures couldn't be verified because the public key is not avai
  lable: NO_PUBKEY ED65462EC8D5E4C5
E: The repository 'http://http.kali.org/kali kali-rolling InRelease' is
  not signed.
N: Updating from such a repository can't be done securely, and is theref
  ore disabled by default.
N: See apt-secure(8) manpage for repository creation and user configurat
  ion details.
```

```
(tce@TCEDA)-[~]
$ sudo apt install nfs-common -y
[sudo] password for tce:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
nfs-common is already the newest version (1:1.3.4-6).
nfs-common set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
```

- ❖ Create mount point:

```
sudo mkdir -p /mnt/nfs/sharedfolder
```

```
(tce@TCEDA)-[~]
$ sudo mkdir -p /mnt/nfs/sharedfolder

(tce@TCEDA)-[~]
$
```

❖ Mount the NFS share:

sudo mount ubuntu ip:/srv/nfs/dir_name /mnt/nfs/dir_name

eg:

sudo mount 192.168.56.101:/srv/nfs/sharedfolder /mnt/nfs/sharedfolder

```
(tce@TCEDA)-[~]
$ sudo mount 192.168.56.101:/srv/nfs/sharedfolder /mnt/nfs/sharedfolder

(tce@TCEDA)-[~]
$
```

❖ Verify:

df -h | grep nfs

ls -l /mnt/nfs/sharedfolder

```
(tce@TCEDA)-[~]
$ df -h | grep nfs
192.168.56.101:/srv/nfs/sharedfolder 49G 15G 33G 31% /mnt/nfs/sharedfolder

(tce@TCEDA)-[~]
$ ls -l /mnt/nfs/sharedfolder
total 0
```

5. Test File Transfer

❖ On Ubuntu VM, create file:

echo "Hello from Ubuntu" | sudo tee /srv/nfs/dir_name/ubuntu_test.txt

```
tce@tce-VirtualBox:~$ echo "Hello from Ubuntu" | sudo tee /srv/nfs/sharedfolder/ubuntu_test.txt
Hello from Ubuntu
```

❖ On Kali VM, verify(READ OPERATION)

ls -l /mnt/nfs/sharedfolder

cat /mnt/nfs/sharedfolder/ubuntu_test.txt

```
(tce@TCEDA)-[~]
$ ls -l /mnt/nfs/sharedfolder
total 4
-rw-r--r-- 1 root root 18 Aug 11 14:49 ubuntu_test.txt

(tce@TCEDA)-[~]
$ cat /mnt/nfs/sharedfolder/ubuntu_test.txt
Hello from Ubuntu
```

❖ On Kali VM, (WRITE OPERATION):

```
(tce@TCEDA)-[~]
$ echo "Hello from Kali" | sudo tee /mnt/nfs/sharedfolder/kali_test.tx
t
Hello from Kali
```

❖ On Ubuntu VM, verify

```
ls -l /srv/nfs/sharedfolder
```

```
cat /srv/nfs/sharedfolder/kali_test.txt
```

```
tce@tce-VirtualBox:~$ cat /srv/nfs/sharedfolder/kali_test.txt
Hello from Kali
tce@tce-VirtualBox:~$
```

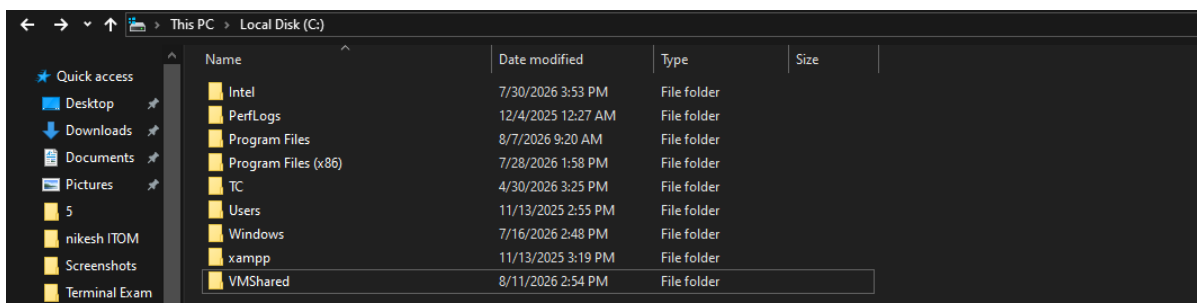
METHOD 2

SHARED FOLDER

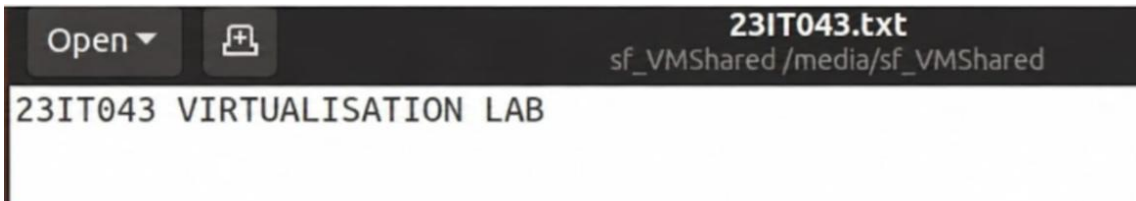
Step 1: Create a Shared Folder on Your Host

1. On your host machine (Windows/Linux):

- Create a folder, e.g., C:\VMShared



- Add a test file (22IT043.txt) to confirm later.

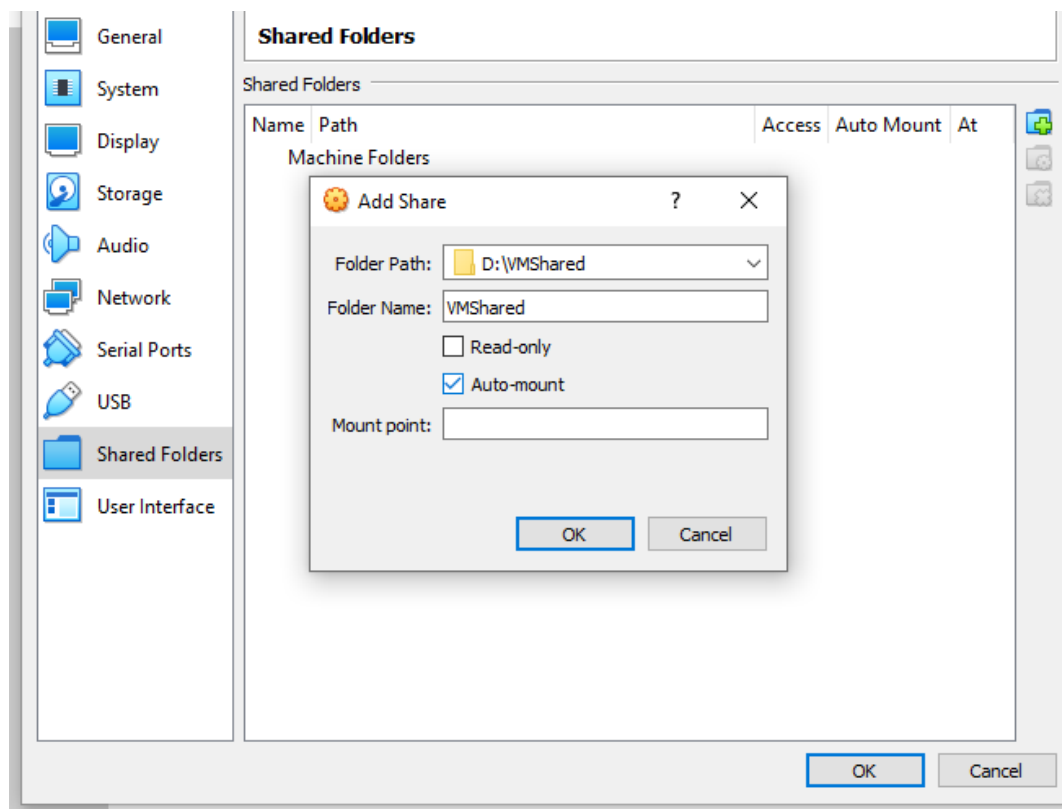


Step 2: Enable Shared Folder in VirtualBox (for Both VMs)

Repeat this for both Ubuntu VMs.

!Discard the saved state before doing it.

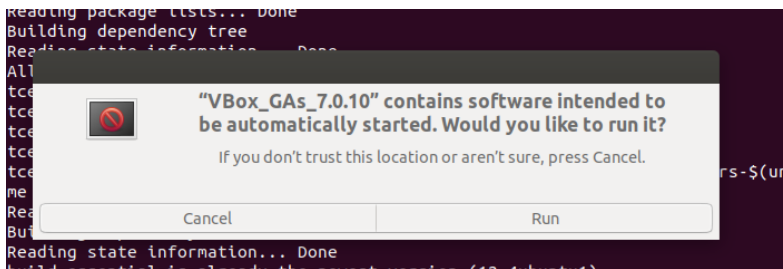
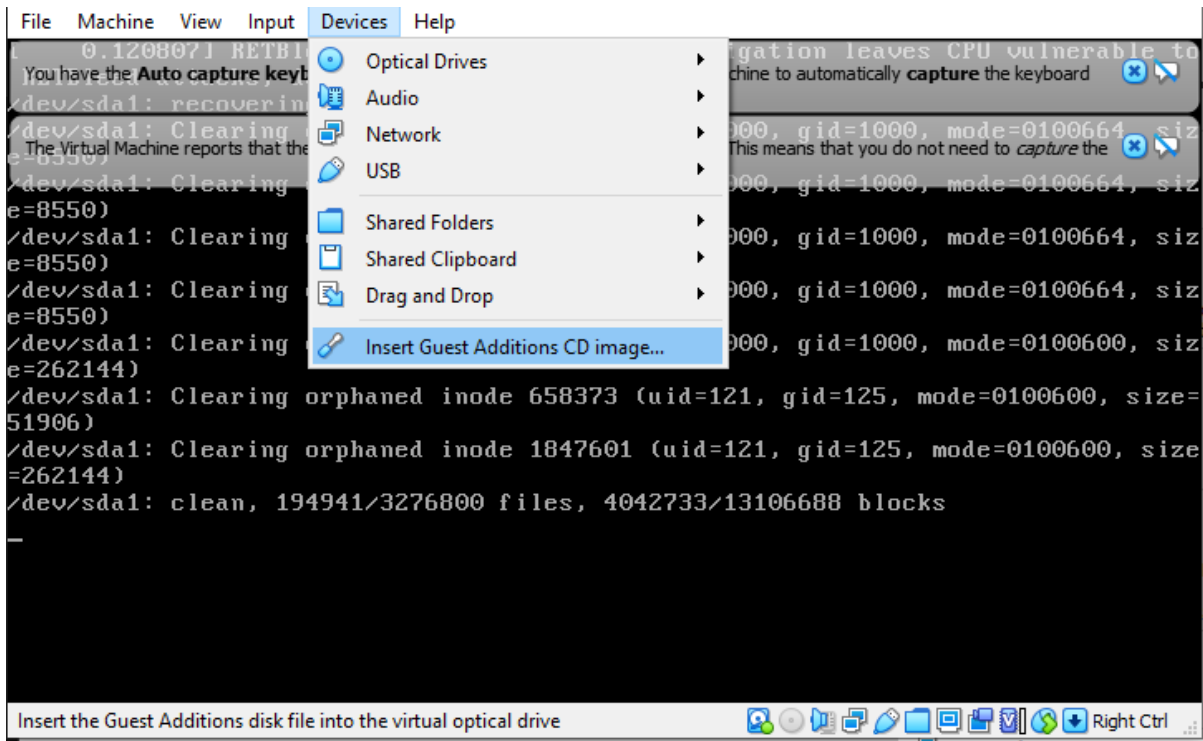
1. Shut down the VM if it's running.
2. In VirtualBox Manager, select your VM.
3. Click Settings → Go to Shared Folders tab.
4. Click “+” (Add new shared folder).
5. Set:
 - Folder Path: Select C:\VMShared (or wherever you created it).
 - Folder Name: VMShared (updates once path is entered)
 - Check(Tick) Auto-mount



Click OK.

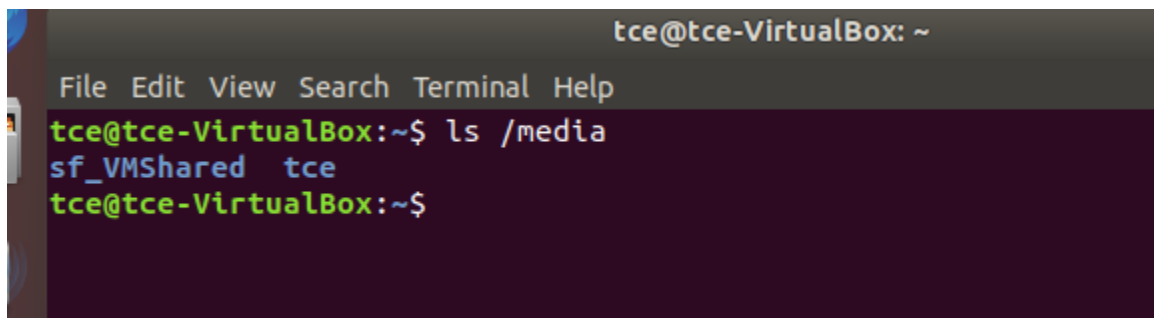
6. Then, again Click OK (at bottom) to save.

3.In VirtualBox menu (inside VM window), go to: *Devices* → *Insert Guest Additions CD image*
VM1:



Now use the command: `ls /media`

- It lists all the directories (i.e., mounted devices or drives) under the /media directory.



Therefore, in VM1 `sf_VMShared` folder is there in `/media` directory.

VM2: SAME PROCEDURE FOR virtualbox 2

Run the command: ls /media

```
ubuntu@ubuntu1804:~$ ls /media
sf_VMShared  ubuntu
ubuntu@ubuntu1804:~$
```

Therefore, in VM2 also sf_VMShared folder is there in /media directory.

Step 4: Access the Shared Folder

After rebooting each VM, the shared folder should be mounted automatically at:
/media/sf_VMShared

To check, run the command : ls /media

VM1 :

```
tce@tce-VirtualBox:~$ ls /media/
sf_VMShared  tce
tce@tce-VirtualBox:~$
```

VM2 :

```
ubuntu@ubuntu1804:~$ ls /media
sf_VMShared  ubuntu
ubuntu@ubuntu1804:~$
```

Run the command : ls /media/sf_VMShared

VM1 :

```
tce@tce-VirtualBox:~$ ls /media/sf_VMShared
ls: cannot open directory '/media/sf_VMShared': Permission denied
tce@tce-VirtualBox:~$
```

VM2 :

```
ubuntu@ubuntu1804:~$ ls /media/sf_VMShared
ls: cannot open directory '/media/sf_VMShared': Permission denied
```

You should see the 22IT043.txt file from your host.

But in VM1 and VM2 , it asks for permission to list the files of sf_VMShared folder.

Step 5: Fix Permissions (if Permission Denied)

You need to add your user to the vboxsf group:

sudo usermod -aG vboxsf \$USER

```
tce@tce-VirtualBox:~$ sudo usermod -aG vboxsf $USER
[sudo] password for tce:
tce@tce-VirtualBox:~$
```

```
ubuntu@ubuntu1804:~$ sudo usermod -aG vboxsf $USER
```

Then reboot again:

sudo reboot

Now try again:

ls /media/sf_VMShared``

```
tce@tce-VirtualBox:~$ ls /media/sf_VMShared
23IT126.txt
tce@tce-VirtualBox:~$
```

```
(tce@TCEDA)-[~]
$ ls /media/sf_VMShared
23IT126.txt
```

You should now have full access.

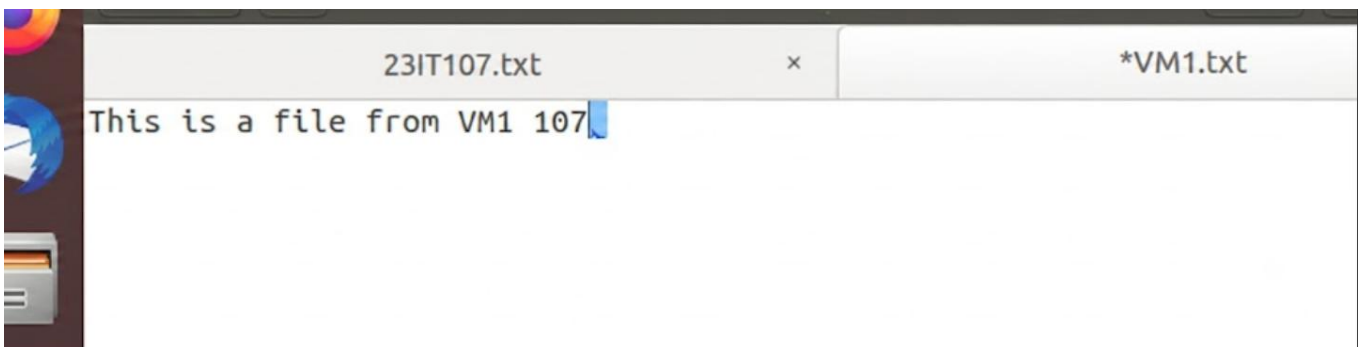
Step 6: Transfer Files Between VMs

- Trying to print the content of 23IT107.txt file from VM1



- Now in VM1, copy a file to /media/sf_VMShared folder,

```
tce@tce-VirtualBox:~$ touch VM1.txt
tce@tce-VirtualBox:~$ gedit VM1.txt
tce@tce-VirtualBox:~$
```



cp VM1.txt /media/sf_VMShared/

- Switch to VM2, and check the shared folder:

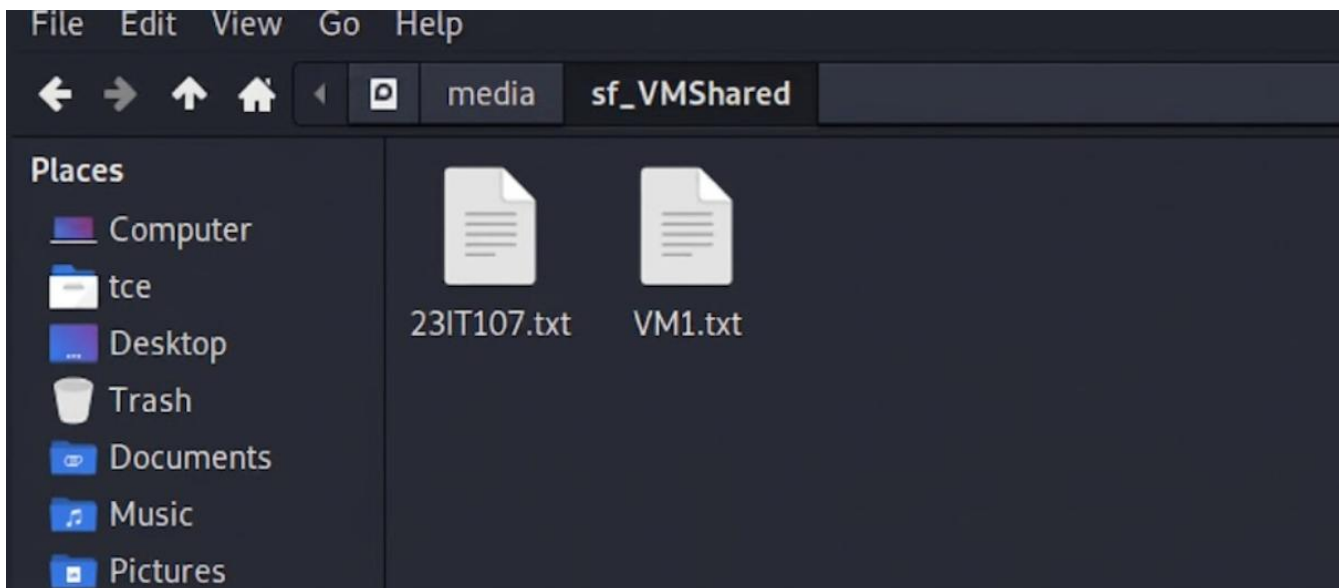
ls /media/sf_VMShared

The VM1.txt file will appear — you can copy it out to any other location and also you can edit files in shared folder location itself. Any changes done in shared folder using one VM will also be reflected to the other VM.

```
(tce@TCEDA)-[~]  
$ ls /media/sf VMShared  
23IT126.txt VM1.txt
```

```
(tce@TCEDA)-[~]  
$ cat /media/sf__VMShared/VM1.txt  
This is a file from VM1 107
```

Also 23IT107.txt file is accessible in VM2,



METHOD 3(SSH Secure Shell)

Step 1: Install SSH

On both VMs (VM1 & VM2):

sudo apt update

sudo apt install openssh-server -y

```
tce@tce-VirtualBox:~$ sudo apt install openssh-server -y
[sudo] password for tce:
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  ncurses-term openssh-client openssh-sftp-server ssh-import-id
Suggested packages:
  keychain libpam-ssh monkeysphere ssh-askpass molly-guard rssh
```

```
(tce@TCEDA)-[~]
$ sudo apt install openssh-server -y
[sudo] password for tce:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
openssh-server is already the newest version (1:8.4p1-5).
openssh-server set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
```

sudo

Check SSH status:

sudo systemctl status ssh

```
tce@tce-VirtualBox:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: ena
   Active: active (running) since Tue 2026-08-11 15:20:04 IST; 56s ago
     Main PID: 2756 (sshd)
        Tasks: 1 (limit: 1108)
       CGroup: /system.slice/ssh.service
              └─2756 /usr/sbin/sshd -D

Aug 11 15:20:04 tce-VirtualBox systemd[1]: Starting OpenBSD Secure Shell server
Aug 11 15:20:04 tce-VirtualBox sshd[2756]: Server listening on 0.0.0.0 port 22.
Aug 11 15:20:04 tce-VirtualBox sshd[2756]: Server listening on :: port 22.
Aug 11 15:20:04 tce-VirtualBox systemd[1]: Started OpenBSD Secure Shell server.
lines 1-12/12 (END)
```

```
(tce@TCEDA)-[~]
$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; disabled; vendor >
   Active: inactive (dead)
     Docs: man:sshd(8)
           man:sshd_config(5)
lines 1-5/5 (END)
```

Step 2: Get Server's IP

On VM1 (Server, username = tce):*ip a*

```
loopback enp0s3: <BROADCAST>
link/ether 08:00:27:a1:92:af brd ff:ff:ff:ff:ff:ff
inet 192.168.56.101/24 brd 192.168.56.255 scope global dynamic noprefixrou
e enp0s3
```

Allow firewall settings for SSH to avoid Connection timed out error.

`sudo ufw enable -- > to enable firewall`

`sudo ufw allow 22/tcp`

`sudo ufw reload`

```
tce@tce-VirtualBox:~$ sudo ufw allow 22/tcp
Rules updated
Rules updated (v6)
tce@tce-VirtualBox:~$ sudo ufw reload
Firewall not enabled (skipping reload)
tce@tce-VirtualBox:~$ sudo enable firewall
sudo: enable: command not found
tce@tce-VirtualBox:~$ sudo ufw enable
Firewall is active and enabled on system startup
tce@tce-VirtualBox:~$ sudo ufw reload
Firewall reloaded
```

Step 3: Connect from VM2 to VM1

On VM2 (Client, username = ubuntu):

`ssh tce@192.168.56.101`

```
└─$ ssh tce@192.168.56.101 130 x
The authenticity of host '192.168.56.101 (192.168.56.101)' can't be established.
ECDSA key fingerprint is SHA256:ZVFqW/52kkcKPIKY0b/7DQ1WtTrv04I2g5Qk4ZYP04Q.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '192.168.56.101' (ECDSA) to the list of known hosts.
tce@192.168.56.101's password:
Welcome to Ubuntu 18.04.6 LTS (GNU/Linux 5.4.0-150-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

58 updates can be applied immediately.
4 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by applicable law.

Your Hardware Enablement Stack (HWE) is supported until April 2023.

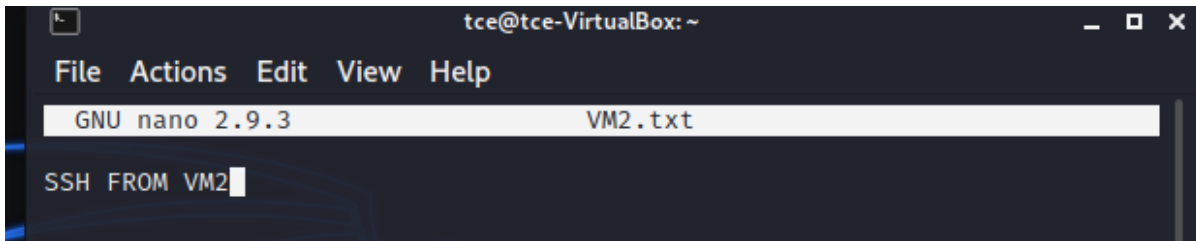
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
```

Enter tce's password → now you are logged into VM1.

Step 4: Transfer Files with SCP (From VM2)**(a) Copy file from VM2 → VM1**

scp file.txt tce@192.168.56.103:/home/tce/

```
tce@tce-VirtualBox:~$ touch VM2.txt
tce@tce-VirtualBox:~$ nano VM2.txt
```

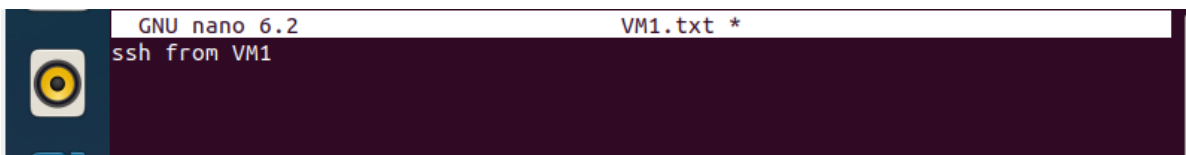


```
tce@tce-VirtualBox:~$ scp VM2.txt tce@192.168.56.101:/home/tce/
The authenticity of host '192.168.56.101 (192.168.56.101)' can't be esta
blished.
ECDSA key fingerprint is SHA256:ZVFqW/52kkcKPIKY0b/7DQ1WtTrv04I2g5Qk4ZYP
04Q.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added '192.168.56.101' (ECDSA) to the list of known
hosts.
tce@192.168.56.101's password:
VM2.txt                                100% 13    16.4KB/s   00:00
```

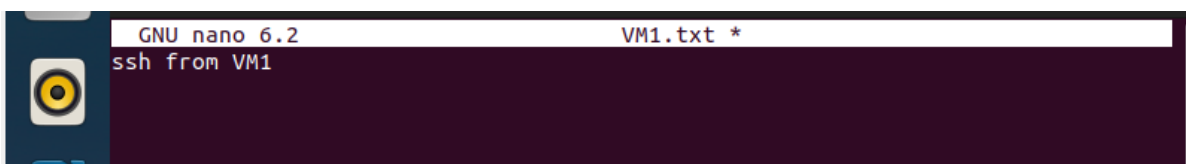
```
tce@tce-VirtualBox:~$ ls
bhar.java  Documents      GeeksforGeeks  Pictures  sum.class  Videos
Desktop    Downloads      itom.txt       Public    Templates  VM1.txt
divs       examples.desktop Music          rithika  test.txt   VM2.txt
tce@tce-VirtualBox:~$ cat VM2.txt
SSH FROM VM2
tce@tce-VirtualBox:~$
```

(b) Copy file from VM1 → VM2

scp tce@192.168.56.103:/home/tce/serverfile.txt /home/ubuntu/



```
ubuntu@ubuntu1804:~$ scp tce@192.168.56.103:/home/tce/VM1.txt /home/ubuntu/
tce@192.168.56.103's password:
VM1.txt                                100% 13    5.5KB/s   00:00
```



METHOD 4(FTP (File Transfer Protocol))

Step 1: Install FTP Server on VM1

(VM1 = server, VM2 = client)

On VM1:

sudo apt update

sudo apt install vsftpd -y

```
tce@tce-VirtualBox:~$ sudo apt install vsftpd -y
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following NEW packages will be installed:
  vsftpd
0 upgraded, 1 newly installed, 0 to remove and 57 not upgraded.
Need to get 115 kB of archives.
```

sudo

Enable and start the service:

sudo systemctl enable vsftpd

sudo systemctl start vsftpd

```
tce@tce-VirtualBox:~$ sudo systemctl enable vsftpd
Synchronizing state of vsftpd.service with SysV service script with /lib/systemd/sd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable vsftpd
tce@tce-VirtualBox:~$ sudo systemctl start vsftpd
```

Step 2: Configure vsftpd

Edit the config file:

sudo nano /etc/vsftpd.conf

```
tce@tce-VirtualBox:~$ sudo nano /etc/vsftpd.conf
[sudo] password for tce:
```

Make sure these lines are set (uncomment or add if missing):

local_enable=YES

write_enable=YES

chroot_local_user=YES

allow_writeable_chroot=YES

Save (CTRL+S, CTRL+X)

```
tce@tce-VirtualBox:~$ grep -v "^#" /etc/vsftpd.conf
listen=NO
listen_ipv6=YES
anonymous_enable=NO
local_enable=YES
write_enable=YES
dirmessage_enable=YES
use_localtime=YES
xferlog_enable=YES
connect_from_port_20=YES
chroot_local_user=YES
secure_chroot_dir=/var/run/vsftpd/empty
pam_service_name=vsftpd
rsa_cert_file=/etc/ssl/certs/ssl-cert-snakeoil.pem
rsa_private_key_file=/etc/ssl/private/ssl-cert-snakeoil.key
ssl_enable=NO
allow_writeable_chroot=YES
```

Restart the ftp server:

```
sudo systemctl restart vsftpd
```

```
tce@tce-VirtualBox:~$ sudo systemctl restart vsftpd
```

Step 3: Check VM1 IP Address

On VM1, run:

```
ip a
```

```
inet 192.168.56.103/24 brd 192.168.56.255 scope global dynamic noprefixroute enp0s3
    valid_lft 364sec preferred_lft 364sec
```

Allow firewall settings for FTP to avoid Connection timed out error.

```
sudo ufw allow 21/tcp
```

```
sudo ufw allow 20/tcp # (for data transfer)
```

```
sudo ufw status
```

```
tce@tce-VirtualBox:~$ sudo ufw allow 21/tcp
Rule added
Rule added (v6)
tce@tce-VirtualBox:~$ sudo ufw allow 20/tcp
Rule added
Rule added (v6)
```

```
tce@tce-VirtualBox:~$ sudo ufw reload
Firewall reloaded
```

Step 4: Connect from VM2 (FTP client)**On VM2, install ftp client:***sudo apt install ftp -y*

```
(tce@TCEDA)-[~]
$ sudo apt install ftp -y
[sudo] password for tce:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ftp is already the newest version (0.17-34.1.1).
ftp set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
```

Now connect:*ftp 192.168.56.103*

Enter VM1's username (tce) and password when prompted.

```
(tce@TCEDA)-[~]
$ ftp 192.168.56.101
Connected to 192.168.56.101.
220 (vsFTPd 3.0.3)
Name (192.168.56.101:tce): tce
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp>
```

Step 5: FTP Commands

Once connected, you can use:

- *ls* → list files on VM1

```
ftp> ls
200 PORT command successful. Consider using PASV.
150 Here comes the directory listing.
drwxr-xr-x  2 1000    1000          4096 May 12  2022 Desktop
drwxr-xr-x  2 1000    1000          4096 May 12  2022 Documents
drwxr-xr-x  3 1000    1000          4096 Aug 05 11:21 Downloads
drwxrwxr-x  2 1000    1000          4096 Aug 07 11:23 GeeksforGeeks
drwxr-xr-x  2 1000    1000          4096 May 12  2022 Music
drwxr-xr-x  2 1000    1000          4096 Aug 11 11:33 Pictures
drwxr-xr-x  2 1000    1000          4096 May 12  2022 Public
drwxr-xr-x  2 1000    1000          4096 May 12  2022 Templates
-rw-rw-r--  1 1000    1000           18 Aug 11 15:32 VM1.txt
-rw-rw-r--  1 1000    1000           13 Aug 11 15:28 VM2.txt
drwxr-xr-x  2 1000    1000          4096 May 12  2022 Videos
-rw-rw-r--  1 1000    1000          1077 Aug 05 11:15 bhar.java
-rw-rw-r--  1 1000    1000           0 Aug 05 11:38 divs
-rw-r--r--  1 1000    1000          8980 May 12  2022 examples.desktop
-rw-rw-r--  1 1000    1000           0 Aug 10 11:25 itom.txt
drwxrwxr-x  2 1000    1000          4096 Aug 04 12:11 rithika
-rw-rw-r--  1 1000    1000          1077 Jul 14 14:30 sum.class
-rw-rw-r--  1 1000    1000           0 Aug 07 11:26 test.txt
226 Directory send OK.
ftp>
```

- *cd folder* → change directory on VM1

```
ftp> cd Downloads
250 Directory successfully changed.
ftp> ls
200 PORT command successful. Consider using PASV.
150 Here comes the directory listing.
drwxrwxr-x   2 1000    1000          4096 Aug 05 11:22 bharathi
-rw-rw-r--   1 64055   127          40304640 Aug 04 14:28 cirros-0.5.1-x86
_64-disk.img
-rw-rw-r--   1 1000    1000           23 Jul 07 14:33 f1.txt
-rw-rw-r--   1 1000    1000           22 Jul 07 14:30 f2.txt
-rw-rw-r--   1 64055   127          1697906688 Aug 04 14:46 ubuntu-16.04.7
-desktop-amd64.iso
-rw-rw-r--   1 64055   127          922746880 Aug 04 15:12 ubuntu-16.04.7-
server-amd64.iso
226 Directory send OK.
ftp> █
```

- *get filename* → download file from VM1 → VM2

```
ftp> get VM1.txt
local: VM1.txt remote: VM1.txt
200 PORT command successful. Consider using PASV.
150 Opening BINARY mode data connection for VM1.txt (18 bytes).
226 Transfer complete.
18 bytes received in 0.01 secs (1.7317 kB/s)
ftp> █
```

- *put filename* → upload file from VM2 → VM1

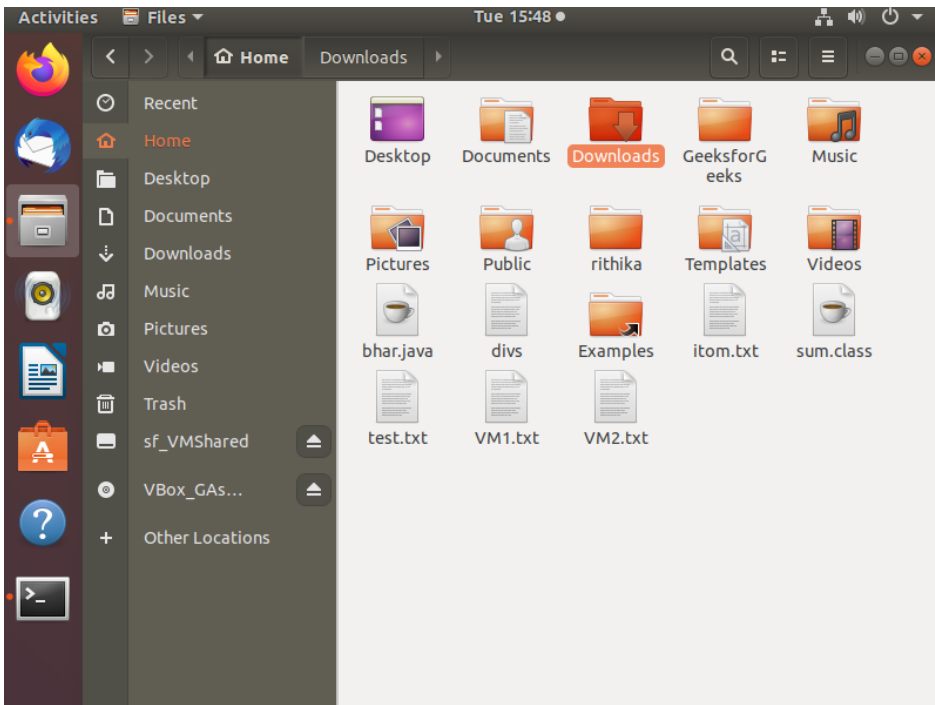
```
ftp> put VM1.txt
local: VM1.txt remote: VM1.txt
200 PORT command successful. Consider using PASV.
150 Ok to send data.
226 Transfer complete.
18 bytes sent in 0.00 secs (254.7554 kB/s)
ftp> █
```

- *bye* → exit

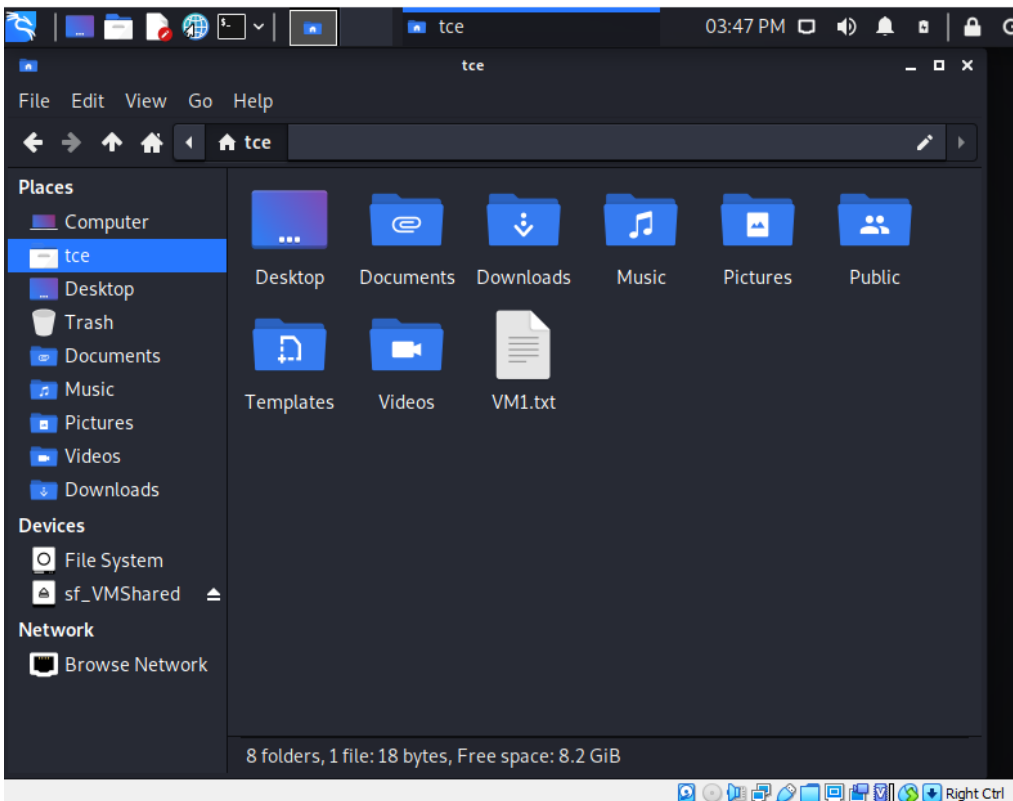
```
18 bytes sent in 0.00 secs (254.7554 kB/s)
ftp> bye
221 Goodbye.
```

Step 6: Verify

- `ftp_vm1.txt` should now be in VM2's current directory.



- **ftp_vm2.txt** should now be uploaded to VM1's **/home/FTP1/** directory.



RESULT:

Thus the file transfer between Ubuntu and Kali VMs is executed successfully and both VMs can read and write files in the shared folder seamlessly.